EC Lyon

Design, manufacturing and characterization of interferometers on the SOI platform

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Contents

Abstract

In this report, we designed and manufactured asymmetric Mach Zehnder as well as Michelson interferometers with Si on SiO2 waveguides. Simulations were carried out using mostly Lumerical and the results were compared to the experimental data obtained. One of the parameters considered was the group index of the fundamental quasi TE mode. Its error at 1550 nm between simulation and experiment was found to be less than 1%.

1 Introduction

In the last years, integrated photonics has become increasingly important and is expected to help facilitate Moores law for the upcoming decades. But already today it has significant contributions, such as in optical communications. Since the semiconductor industry is build on Silicon, it is also the most utilized material for integrated photonics today as it makes it increasingly simple to use the CMOS fabrication processes.

In light of this, the work carried out in this course is devoted to performing all required steps for making devices on photonics platforms. These include design and simulation, layout and fabrication and lastly optical characterization. Here, we use the common platform Silicon on Silicon Dioxide (Silicon-on-insulator SOI) to implement imbalanced interferometers. From the measurements, we will extract the effective as well as the group index and the free spectral range of the transmission spectra.

2 Waveguide

The waveguide used is based on Silicon, with 0.5 microns width and 0.22 microns thickness. The substrate as well as cladding are SiO2. In this configuration, we investigate the fundamental TE mode at around 1550 nm, which is almost entirely guided in the Silicon. We can perform FDE simulations using Lumerical Mode in order to find the spatial field distribution of the desired mode, as shown in figure 1. Since the mode is not entirely confined in the Silicon slab, we need to take into account the surrounding medium in order to find the effective refractive index of the guided mode.

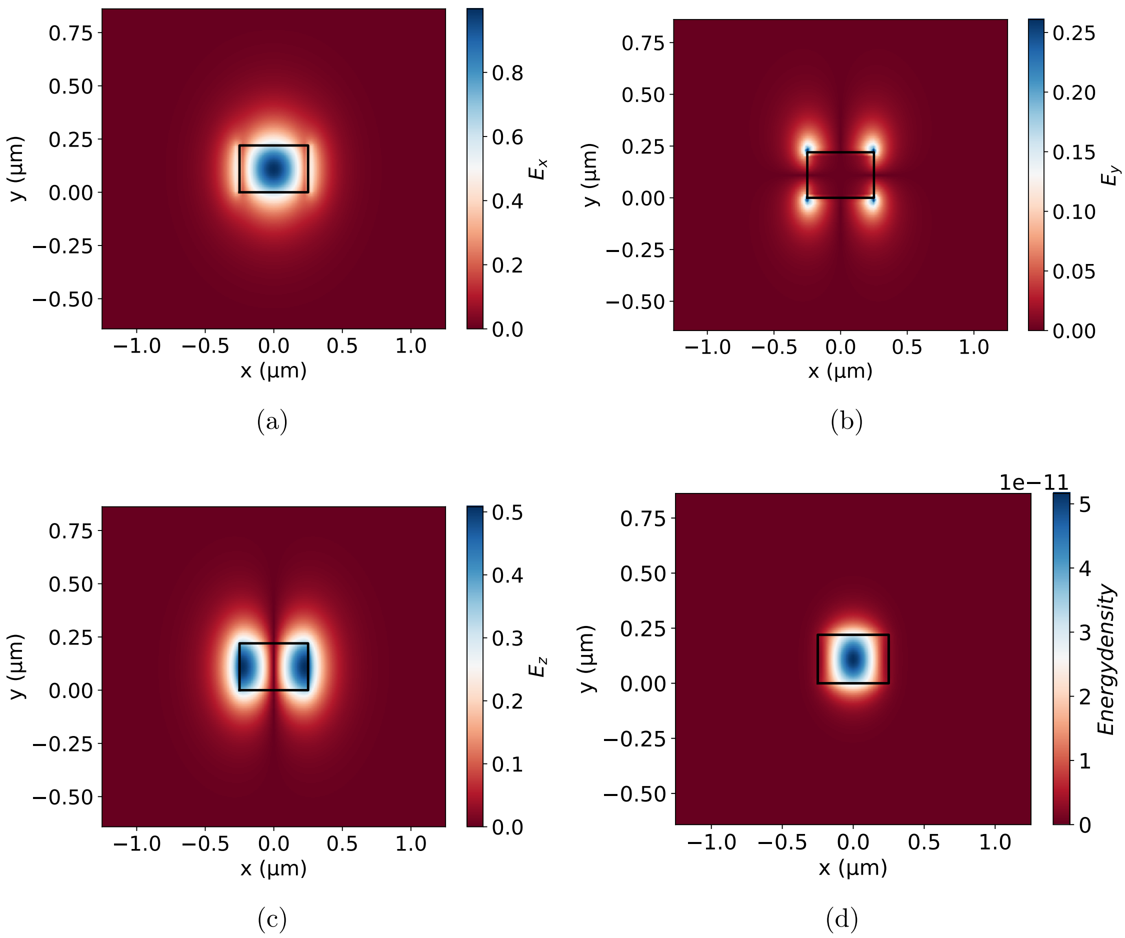


Figure 1: Here, we show the different components of the transverse electric fundamental mode (a, b and c) and the energy density calculated by $W(r) = \frac{1}{2}\left(\epsilon_0|E(r)|^2 + \mu_0|H(r)|^2\right)$. The width of the waveguide is 0.5 μm, the thickness is 0.22 μm

We can also acquire the effective index n_{eff} through the simulation and by changing the frequency and redoing the calculations, find the compact model of the proposed structure. We plot the simulation data as well as fit to the compact model

$$n_{eff} = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2, \tag{1}$$

in figure 2. From the curve of n_{eff} we can infer important parameters to consider for different structures, devices such as dispersion, beta parameters, group velocity, etc.

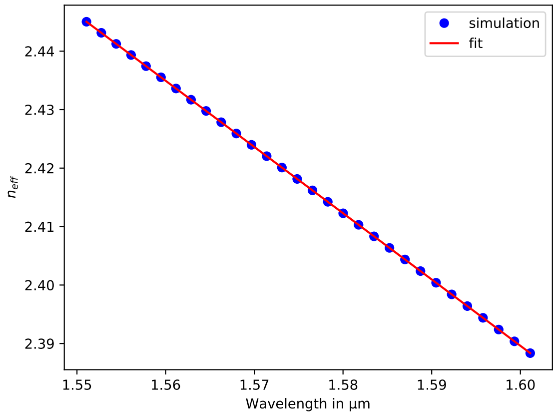


Figure 2: The fit yields $n_{eff} = 2.4462 - 1.1306(\lambda - \lambda_0) - 0.04158(\lambda - \lambda_0)^2$

Knowing the effective index of the guided mode, we can directly import our waveguide model into Lumerical interconnect, which allows us to simulate devices such as interferometers guided through our proposed structures.
Before doing this, it is instructive to think about manufacturing variability. An error introduced is during the thin film deposition by the wafer manufacturer, Soitec reports wafers of thickness 219.2 nm SiO2 thickness plus or minus 3.9 nm. Other errors are introduced in the width of the waveguide due to Lithography and subsequent etching. We take a conservative range from -30 nm to +30 nm.
A very quick way to check how the results change in light of dimension variability is to perform a corner analysis of the parameters that can vary during manufacturing. In our case this will result in 9 different simulations for the different parameter combinations (including the target dimensions). The different curves for effective index as well as group velocity can be seen in figures 3 and 4.

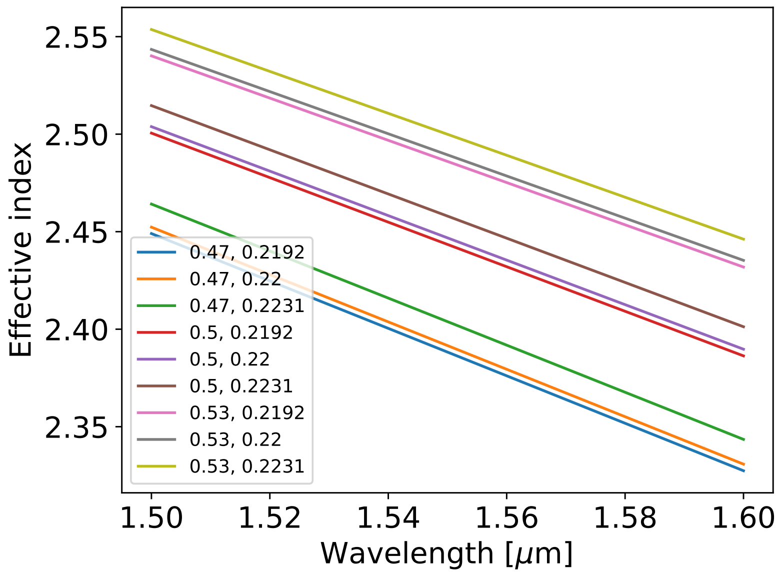


Figure 3: The effective index as calculated from the corner analysis.

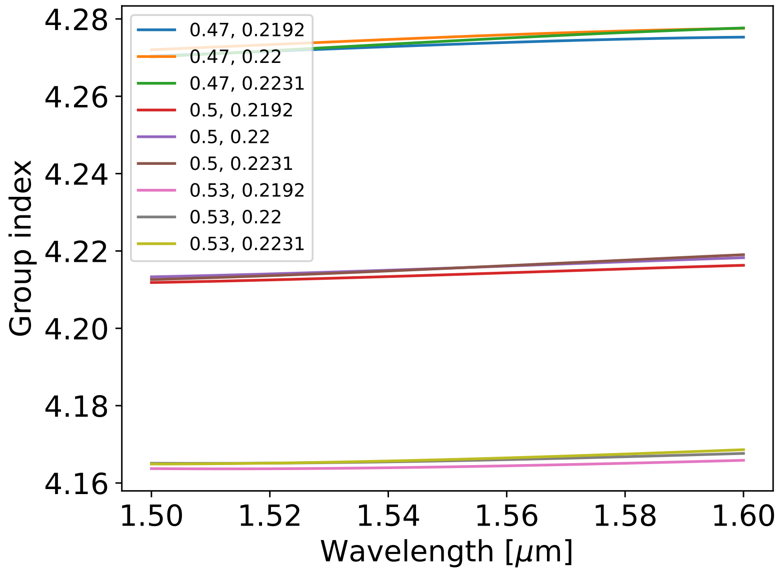


Figure 4: The group index as calculated from the corner analysis.

From the results, we infer that the effective index that we will measure should lie between 2.38 and 2.47 and the group index between 4.17 and 4.27 for the fundamental quasi TE mode at approximately 1550 nm.

3 Design of interferometers

Asymmetric Mach Zehnder interferometers are commonly used to determine the effective index and subsequently the beta parameters of the used platform. Using the waveguide design from Lumerical mode, we can simulate the MZI in interconnect, using the structures from the Ebeam PDK. Namely, grating couplers and Y-branches. The used design can be seen in figure 5. We manufactured slightly different designs, by changing the path length difference, as can be seen in the table below.
The transfer function of the MZI is given by

$$T(\lambda) = 1/2\,(1 + \cos(\beta\Delta L)) \tag{2}$$

and the FSR can be calculated using

$$FSR = \lambda^2/(n_g\Delta L) \tag{3}$$

Table 1: The simulated free spectral range for different arm mismatches ΔL at 1550 nm and for the fundamental quasi TE mode.

FSR	ΔL in μm
0	0
11.3585	50
5.6500	100
2.8905	200

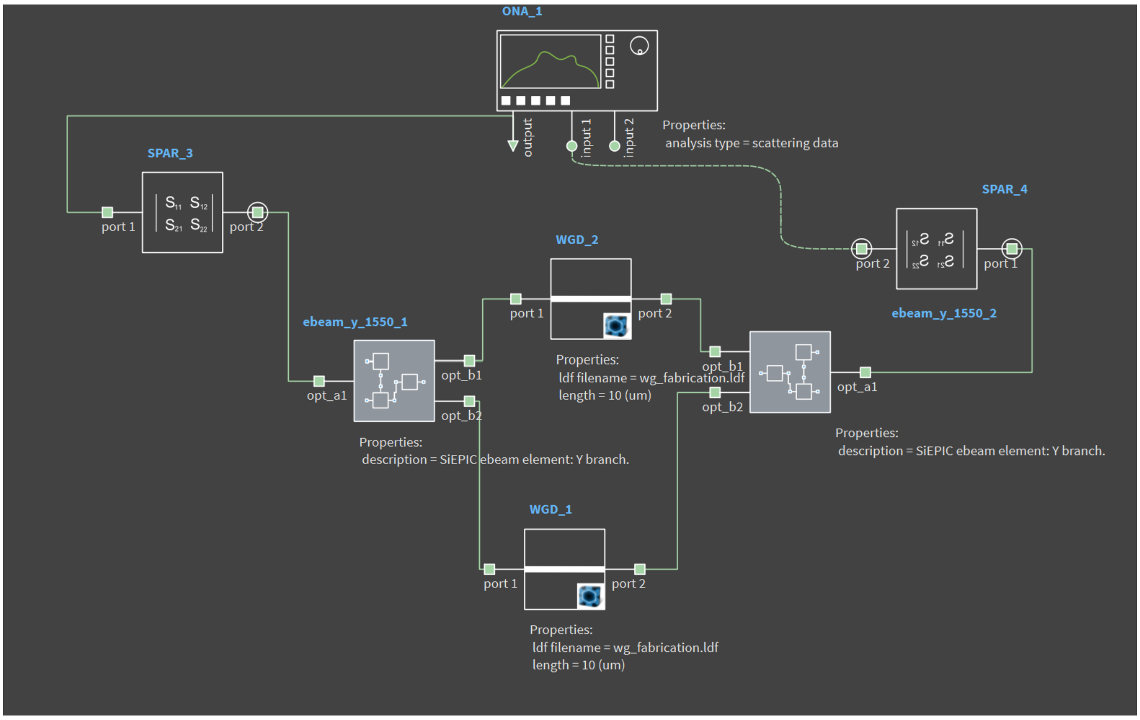


Figure 5: The proposed interferometer, designed in interconnect.

The transfer function is plotted as

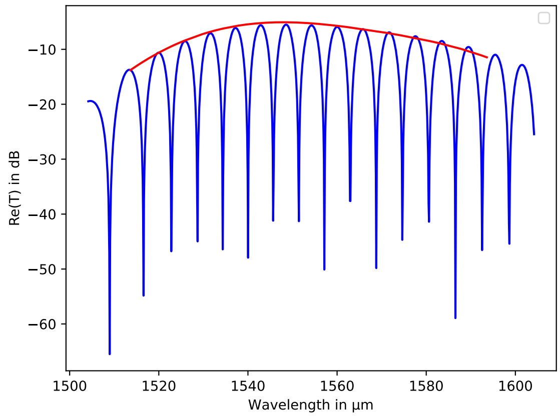


Figure 6: Spectrum of the mach zehnder interferometer with $\Delta L = 100\,\mu$m.

We also modeled a Michelson interferometer using Bragg gratings as frequency selective mirrors as can be seen in figure 7. The used Bragg gratings have the reflection around central wavelength of 1550 nm, so we expect to be able to analyse the transmission data from the measurement.

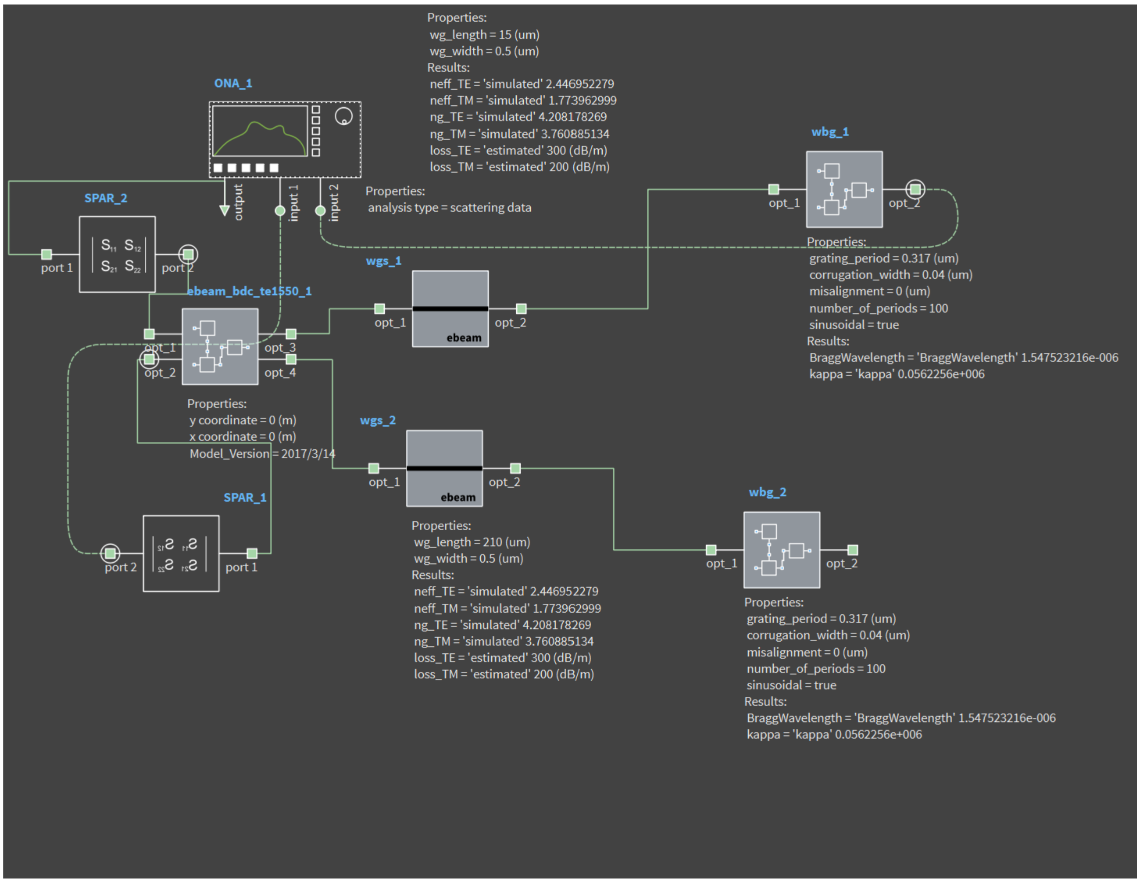


Figure 7: The proposed michelson interferometer using bragg gratings as wavelengthselective mirrors, designed in interconnect.

This gives a spectrum of

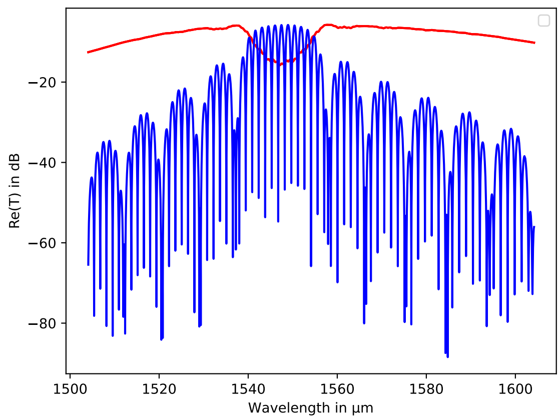


Figure 8: transmission and reflection spectrum of the michelson setup, using bragg gratings as mirrors at center wavelength around 1550 nm.

4 Measurements

The first thing to do is to correct for the finite bandwidth of the grating couplers, for this matter devices using only loop back grating couplers were used. The transmission spectra obtained are displayed in figure 9. The region of interest is between

1.53 and 1.58 microns. In that region a fourth order polynomial is fitted and simply subtracted from all MZ spectra to delete the effects of the grating couplers to our transmission spectra through the interferometer.

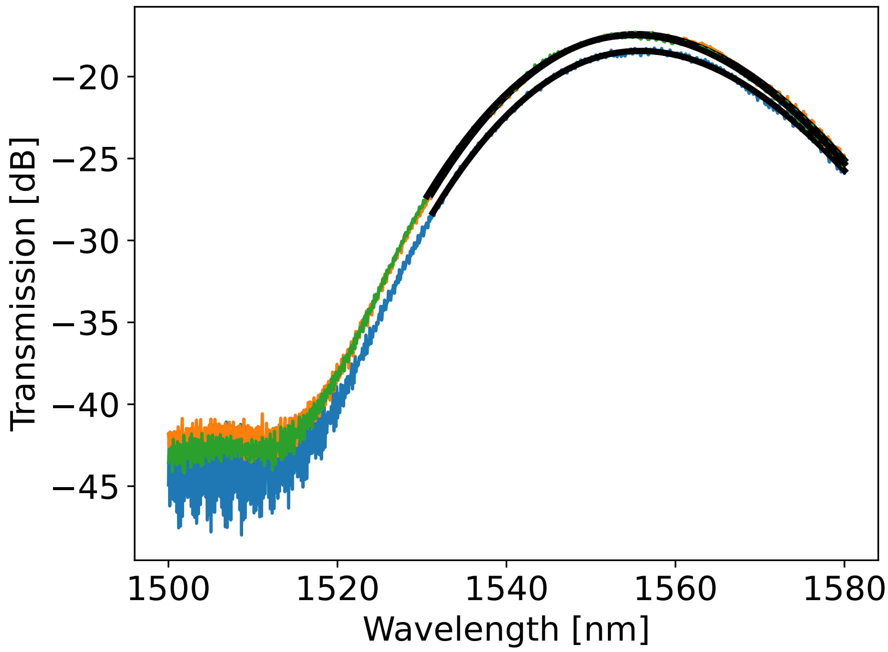


Figure 9: Baseline correction of the grating couplers by measuring the transmission through a loopback system. The black curves correspond to the fitted region of interest with a polynomial up to fourth order.

In figure 10 we display the data for the MZI with the largest mismatch, 400 nm as we expect to get the best result from it since it has many oscillations within our sweep frequency as well as within the accuracy. From the minima of the transmission, we can measure the FSR of our device and then use eq. 3 to calculate the resulting group index.

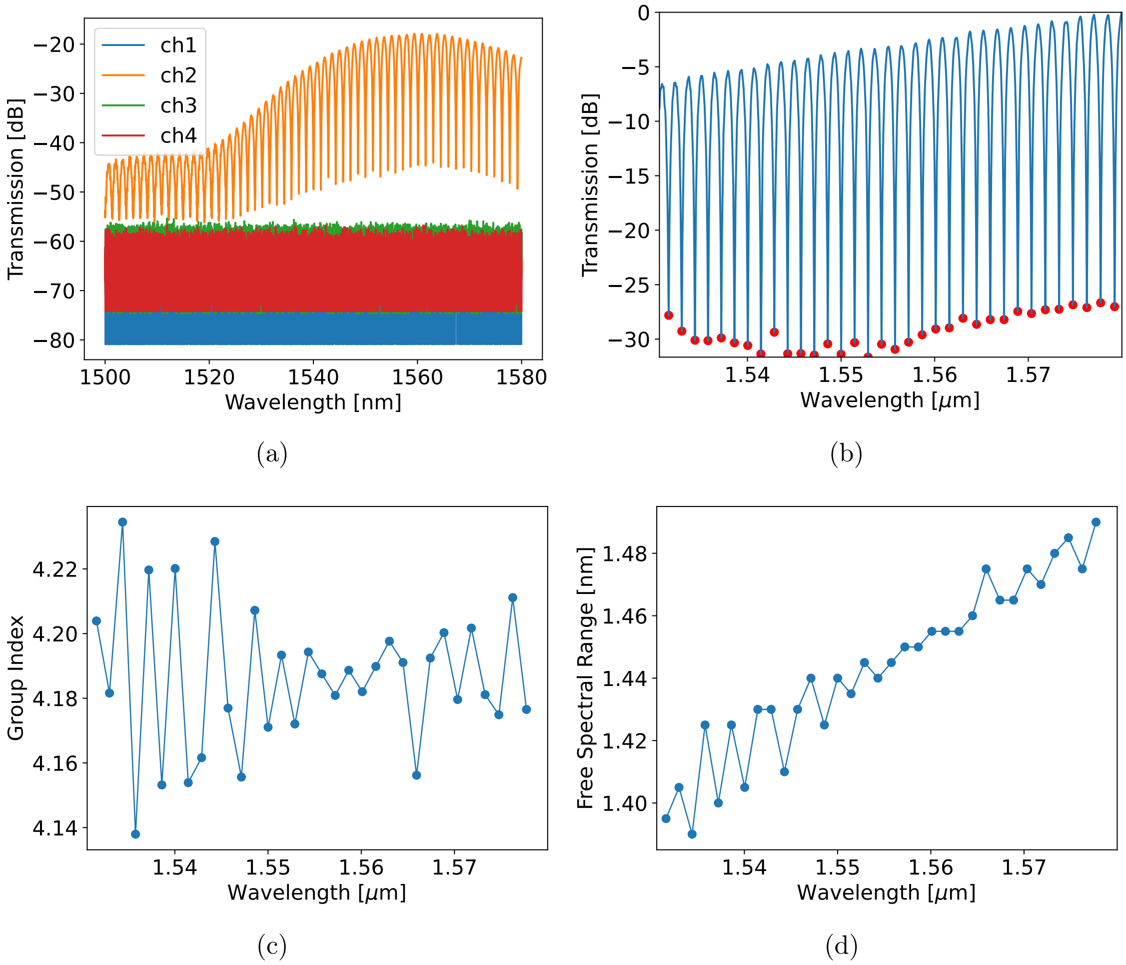


Figure 10: Here, we show the raw data from the experimental run for the MZI with 400 nm mismatch between the arms, the baseline corrected data and the values for FSR as calculated from the destructive interference peaks and the group index.

We can directly obtain all the necessary parameters by fitting the transfer function to the measurement, with a careful consideration of the initial parameters.

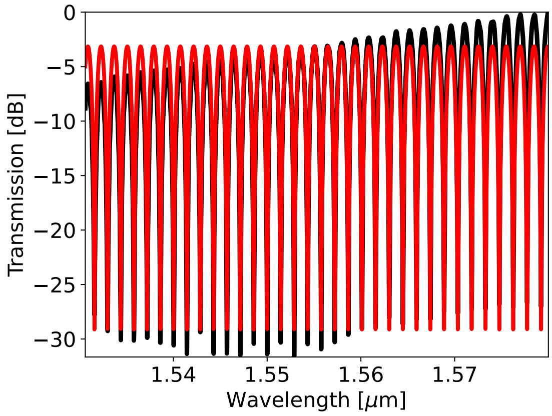
$$F = 10\log_{10}\left(\frac{1}{4}\left|1+\exp\left(-i\frac{2\pi n_{eff}}{\lambda}\Delta L - \alpha\Delta L/2\right)\right|^2\right) + b, \tag{4}$$

where n_{eff} is our compact model given by 1, α is the waveguide loss and b describes any excess insertion losses. Curve fitting was done using a least squares approach in Python, however, since the function is very complex and has 5 variables, this approach only works with a good initial guess. Otherwise, the algorithm will not converge to the optimum. By carefully deciding on the initial parameters and validating the solution "by eye", the analysis was conducted. As can be seen in figure 10(b) the FSR and subsequently n_g were found finding the difference between the minima of transmission in the spectrum. This can be implemented in Python straightforwardly by a minima search algorithm (e.g. "findpeaks" from Scipy).
For the effective index, we choose an initial guess for $n_1 = 2.4$, as we expect the physical solution to be around this value. We can set n_3 to zero, since the slope of

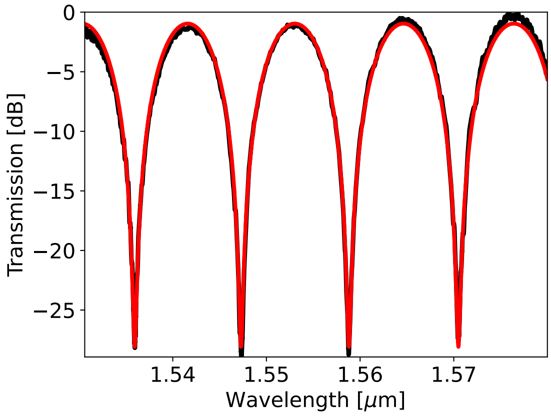
the group index is very small and then relate n_2 to n_1 and the group index as

$$n_2 = -\frac{n_g - n_1}{\lambda_0}. \tag{5}$$

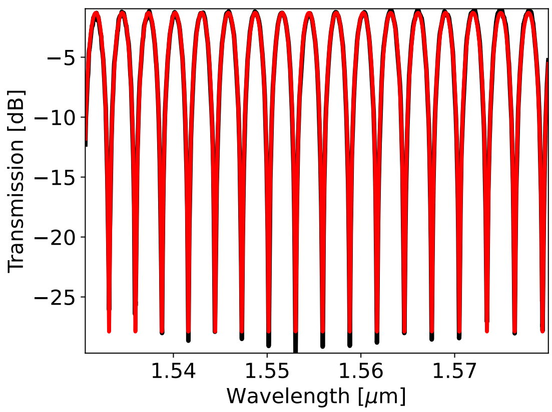
With this we acquire all needed solutions, as displayed in figure 11.



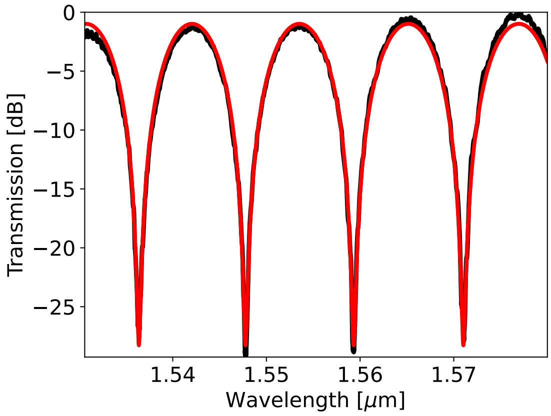
(a) $\Delta L = 400$ nm



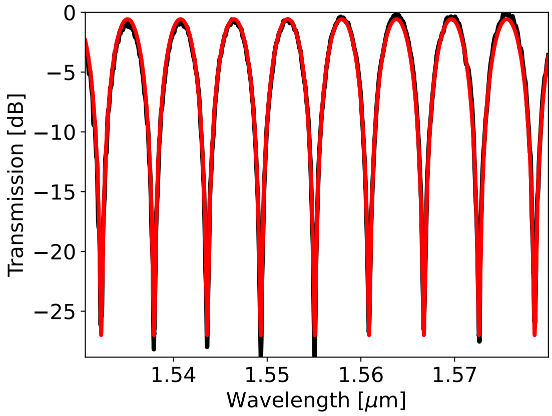
(b) $\Delta L = 50$ nm



(c) $\Delta L = 200$ nm



(d) $\Delta L = 50$ nm



(e) $\Delta L = 100$ nm

Figure 11: The experimental data for the sensitive region (black curves) of the different MZI devices and the fitted function (red curves). Note, we made two devices with mismatch of $\Delta L = 50$ nm.

And the fitted parameters can be seen in the table below

Table 2: Fitted parameters for the different MZI devices.

Name	ΔL [nm]	n_1	n_2	n_3	α	b	R^2
MZI1	400	2.402	-1.149	-5.15e-02	5.028e-04	-2.72	0.922
MZI2.2	50	2.385	-1.156	0.0165	0.00353	-0.587	0.995
MZI2.3	50	2.386	-1.152	-0.0727	0.00346	-0.614	0.994
MZI3	100	2.403	-1.146	-5.20e-03	1.91e-03	-1.64e-01	0.996
MZI4	200	2.400	-1.150	-3.27e-02	9.37e-04	-8.99e-01	0.998

Interestingly, we can plot the function for the effective index and compare the result to the corner analysis done earlier, figure 3. We display the experimentally obtained curves in figure 12. From the comparison, we see that we are within the confidence interval as given by the corner analysis.

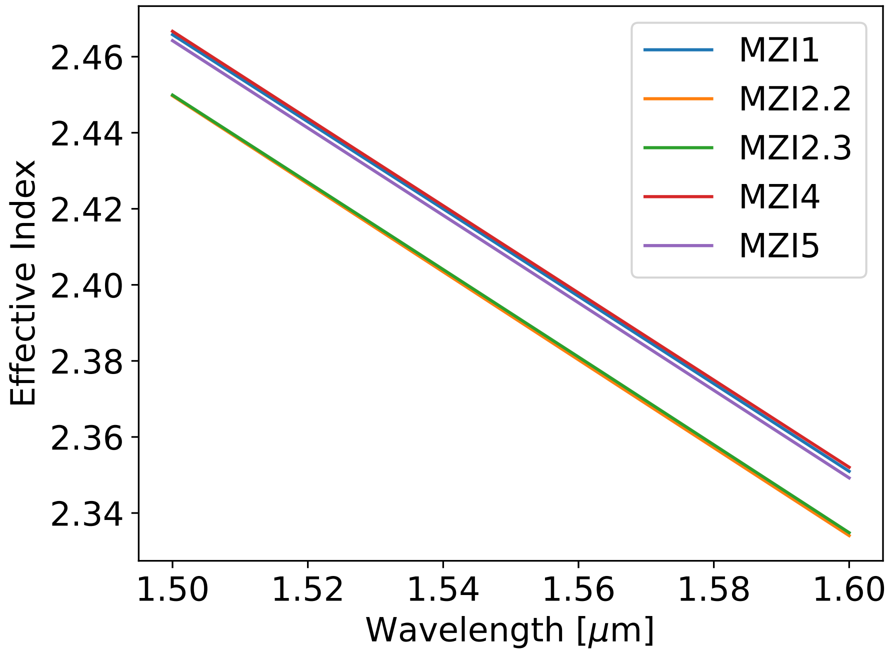


Figure 12: The effective index curves obtained from the measurement data.

And the group index can be found using the relation

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda \frac{dn_{eff}}{d\lambda}, \tag{6}$$

the resulting functions for the group index are shown in figure 13.

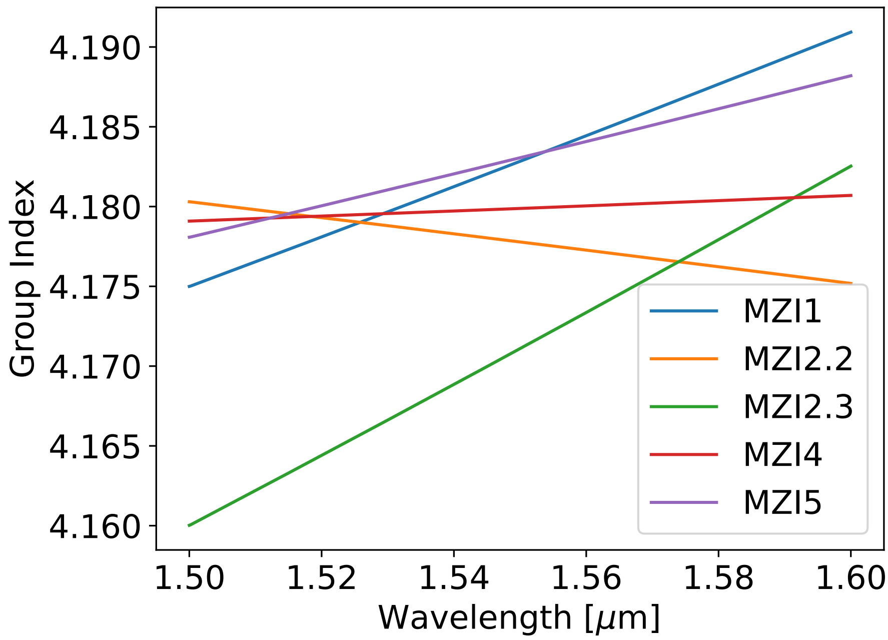
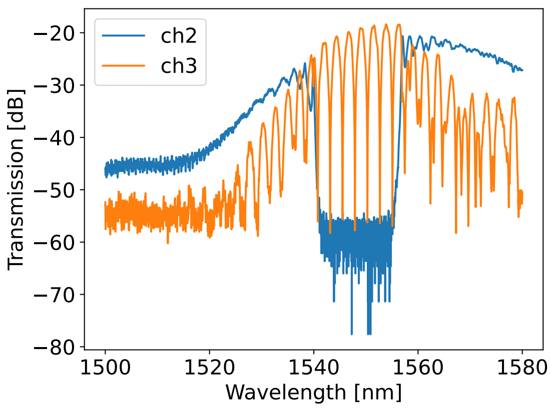
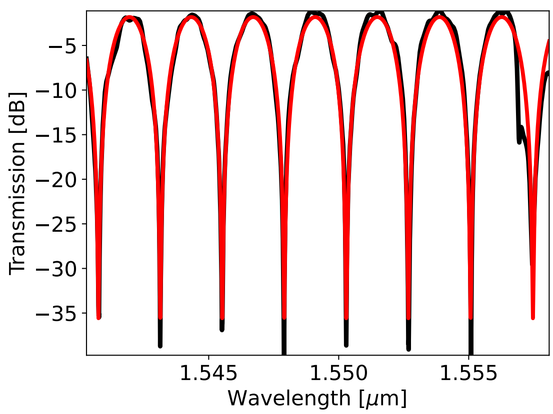


Figure 13: The group index curves obtained from the measurement data.

On top of the MZIs, a Michelson interferometer was designed using Bragg gratings. In principle, the same transfer function can be used if the Y-branches remain 50/50 splitters and by noting that since the light is reflected back and combined, the path length difference is $\Delta L_{true} = 2\Delta L$.



(a)



(b)

Figure 14: Here, we show the raw data from the experimental run for the MMI with total 240 nm mismatch between the arms and the final result for the fitted transfer function.

The FSR at 1550 nm is approximately 2.398 nm which is close to what we expect for an interferometer with mismatch $\Delta L = 240$ nm.

5 Conclusion

We have performed a first estimate of the performance of different devices on the SiO2/Si/SiO2 platform. The interplay of simulation and manufacturing allows us

to optimize the performance of the devices and usually when dealing with a new platform and/or new devices, multiple iterations of simulation/manufacturing/characterization have to be done. Here, we extracted important parameters of the designed interferometers experimentally and compared them to our initial simulations. Namely, the effective index as well as group index of different structures. Firstly, we calibrated the obtained spectra, by using a loop back device in order to delete the insertion loss introduced by the grating couplers. The calibrated spectra were then fitted to corresponding transfer functions of the interferometer. The average group index of the manufactured devices was found to be approximately 4.18 and is very close to the value obtained from the FDE simulations at 4.21.
For a better analysis, the actual loss of the waveguides as well as bending losses should have been experimentally determined, by e.g. using spiral structures. Also, the effect of the grating couplers was not entirely subtracted from the spectra, as can be seen clearly in the data from the MZI with $\Delta L = 400$ nm, which can be optimized by placing the grating couplers closer to the device on the chip. Lastly, more time could have been spend on the characterization of the Bragg gratings of the designed Michelson interferometer.

6 Acknowledgments

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